

Weeks 108 Acoustic Treatment

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1 Introduction

This article serves as a step-by-step guide through the process of measuring and treating a small room at the University of Miami Frost School of Music. This article will cover microphone and speaker equipment, audio interfaces, software, equations, and acoustic panels and treatment.

2 Topics of Interest

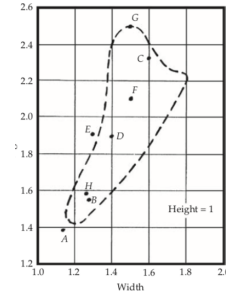
1. Equipment and Software
2. Room Dimensions and Bolt's Area
3. Schroeder Cutoff Frequency
4. Modal Density and Bonello's Criterion
5. Axial Modes of the Room
6. RT60 at 125 Hz, 250 Hz, 500 Hz, and 1000 Hz
7. Treatment Plan

3 Equipment and Software

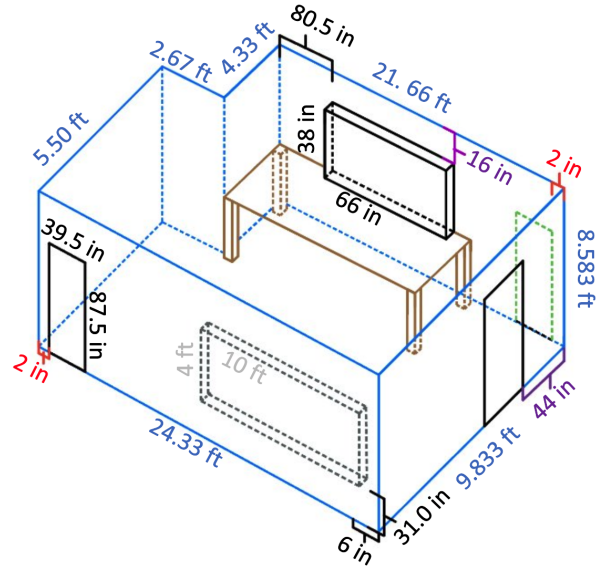
The equipment and software used for this project were as follows:

1. Earthworks M50s
2. Focusrite Scarlet
3. Cables
4. Tape Measure
5. Tripod
6. QSC K12.2 2KW 75° Active Loudspeaker
7. Room EQ Wizard (REW)
8. NIOSH SLM
9. Laser Pointer
10. Small Mirror

4 Room Dimensions and Bolt's Area



4.1 Room Dimensions



Weeks 108 is not a perfectly rectangular room, it has an indent in the southeast edge of the room. Here are the room dimensions, which were measured using a tape measure. The Length (L), Width (W), Height (H), Length of Indent (LI), and Width of Indent (WI) are given as follows:

$$L = 24.33 \text{ ft}$$

$$W = 9.833 \text{ ft}$$

$$H = 8.583 \text{ ft}$$

$$LI = 2.667 \text{ ft}$$

$$WI = 4.333 \text{ ft}$$

This means the Floor-plan Area (FPA) is as follows:

$$FPA = (L \cdot W) - (LI \cdot WI) = 227.7 \text{ ft}^2$$

For purposes discussed in the next section, the floor plan area will be calculated as if the indent was not present. We will call this floor plan area the Hypothetical Floor plan Area (HFA).

$$HFA = L \cdot W = 239.2 \text{ ft}^2$$

For simplicity later, the volume of the room, V , and the hypothetical volume of the room (volume of the room if the indent was not present), HV , will also be calculated now:

$$V = (L \cdot W \cdot H) - (LI \cdot WI \cdot H) = 1954 \text{ ft}^3$$

$$HV = (L \cdot W \cdot H) = 2053 \text{ ft}^3$$

4.2 Bolt's Area

Bolt's Area is given by a ratio in relation to the height of a room. If the length and width of the room are in certain proportions compared to the height, the room is said to be within Bolt's Area and thus optimal for room acoustics. The room in question is not perfectly rectangular and therefore does not fit within Bolt's Area. However, suppose the room did not have this indent in it. Taking a look at the HFA by only using the length, width, and height of the room, on can produce this ratio:

$$1.00 : 1.15 : 2.83$$

Even with this Hypothetical Floor Plan Area, the length of the studio is too long in comparison to the small height of the room and thus the studio does not fit within Bolt's Area. The ratio that this room closes matches to would be Volkmann's: 1.00:1.5:2.5, however this is still not remotely close to the ratio that is in the Weeks studio.

4.3 Gathering measurements

To measure the frequency response of a room follow this procedure:

1. First set the room to how the room will mostly be used. Get rid of as many objects that would not normally be present in the room, move any desks or furniture to the spot that they will most likely stay.
2. Open up Room EQ Wizard and gather the other materials.
3. Connect and set up the audio interface, speaker, and microphone(s).
4. Set your desired sample rate (either 44.1kHz or 48kHz)
5. Connect the output of the interface into the input of the interface. Then match the level of the output to the input level that REW is generating. This is done to remove any frequency response that is self imposed by the equipment.

6. Upload the calibration file (make sure it is the calibration file for the angle you will be recording at) of the microphone into REW, again this is to avoid the microphones frequency response from imposing itself upon the measurements.
7. Calibrate the Input level by opening up the SPL tool in REW. REW should start playing some noise through the loudspeaker. Using the SPL meter, raise the volume of the loudspeaker until your phone or SPL meter reads at about 80dB SPL.
8. Now the measurements can be taken. The microphone will be moved to different positions in the room; the minimum is 5 positions with 3 measurements at each positions. One of the best methods from capturing the frequency response of the room is to play a slow swept sine from 20Hz to 20kHz. Be sure to save each file with a meaningful name.

5 Schroeder Cutoff Frequency

The Schroeder cutoff frequency relates the RT_{60} , the amount of time it takes for a sound to drop by 60 dB, of the room to the volume of the room. It is important because frequencies below this cutoff are the main frequencies that need to be treated. The equation for the Schroeder cutoff frequency, F_2 , is given by:

$$F_2 = 11,250 \sqrt{\frac{RT_{60}}{V}}$$

Where V is the volume of the room. To calculate the RT_{60} , use this equation:

$$RT_{60} = 0.049 \cdot \frac{V}{A}$$

Where A is given by the Sabine equation:

$$A = S_1\alpha_1 + S_2\alpha_2 + S_3\alpha_3 + \dots$$

Where: A is the total Absorption of the room, S_n is the surface area of a surface in your room and α_n is the absorption coefficient of the surface in question. Now would be a great time to take a look at the interior of the room and determine what each surface consists of in order to get an accurate number for the RT_{60} of the room.



Luckily, REW has a function to view the RT60 at various frequencies. Let us compare the REW Schroeder Cutoff Frequency at 500Hz and see how it differs from the calculation done using the equations above.

The walls was painted gypsum board, the floor was polyester carpet and the ceiling was made of PVC ceiling tiles. . Using some lookup tables that can be found online or can be found at the back of the Master Handbook of Acoustics for Musicians by F. Alton Everest, it can be seen that the absorption coefficient α at 500Hz for the materials go as follows:

$$\begin{aligned} \text{Gypsum} \quad \alpha_g &= 0.05 \\ \text{Carpet:} \quad \alpha_c &= 0.14 \\ \text{Ceiling Tile:} \quad \alpha_{ct} &= 0.02 \end{aligned}$$

Sabine Equation:

$$A = \alpha_g \cdot (S_{\text{north}} + S_{\text{east}} + S_{\text{south}} + S_{\text{west}} + S_{\text{s.eWidth}} + S_{\text{s.eastLength}}) + FPA\alpha_c + FPA\alpha_{ct} = 95.08$$

Putting it all together:

$$RT_{60} = 0.049 \cdot \frac{1954}{95.08} = 1.4s$$

$$F_{\text{equation}} = 11,250 \sqrt{\frac{1.4}{1954}} = 301.1 \text{ Hz}$$

The RT_{60} given by REW at 500 Hz was measured as 0.344 seconds. The REW Schroeder Frequency is given below:

$$F_{REW} = 148.4 \text{ Hz}$$

The calculation and the measured RT60s at 500Hz are not close to one another, but it is best we use the REW measurements as this is actually taking into account many different factors that may not be present to the eye. Without ripping the carpet, there is no way to know what material is under the carpet, likewise the exact material coefficients are not known either. For this reason it is best to use the REW RT_{60} s and Schroeder Frequency.

6 Modal Density and Bonello's Criterion

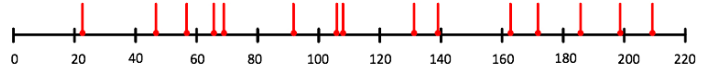
Now, the next step is to list the axial modes up to the Schroeder Frequency. Modes are often indicated as such, (p,q,r) where p represents a mode that exists length-wise in the room, q represents a mode that exists width-wise in the room, and r represents a mode that exists height-wise in the room; p,q, and r will always be integers and thus represent the order of the mode. For example a (2,0,0) mode is a second order mode that exists length-wise in the room. Depending on the number of 0s that are present in-between the parenthesis determines what type of mode being looked at. When two 0s are present, it is an axial mode. When one 0 is present it is a tangential mode, and when no 0s are present it is an oblique mode. To calculate the frequency of a specific mode use this following equation:

$$F = \frac{c}{2} \sqrt{\left(\frac{p^2}{L^2}\right) + \left(\frac{q^2}{W^2}\right) + \left(\frac{r^2}{H^2}\right)} \quad (1)$$

Where c is the speed of sound. Using this equation and only listing the axial modes that produce frequencies that are below 148.4 Hz a chart can be produced:

Mode Number	Integers:			Mode Frequency
	p	q	r	
1	1	0	0	23.222
2	2	0	0	46.444
3	0	1	0	57.459
4	0	0	1	65.827
5	3	0	0	69.666
6	4	0	0	92.889
7	0	2	0	114.919
8	5	0	0	116.111
9	0	0	2	131.655
10	6	0	0	139.334
11	7	0	0	162.556

Axial Modes up to Schroeder Cutoff Frequency

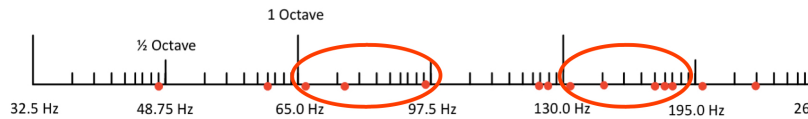


Modal Spacing

These two graphics reveal modal issues with the room. There are many modes where the separation between modes is greater than 20Hz, these modes are: 1 and 2, 5 and 6, 6 and 7, and 10 and 11. There also is a mode that is less than 2 Hz apart, this exists between modes 7 and 8. For all piratical purposes it is best to simply average these two modes together, mode 7 is 114.9 Hz and mode 8 is 116.1 Hz, so the new mode is 115.9 Hz.

6.1 Modal Density

Here is a number line that is in $\log_2$; $\log_2$ is used because the human ear hears in this manner. Simply plotting the modes from 32.5 Hz up to the Schroeder Frequency in this manner allows one to closely look at the more problematic octave bands and will ultimately allow proper for a proper treatment plan.

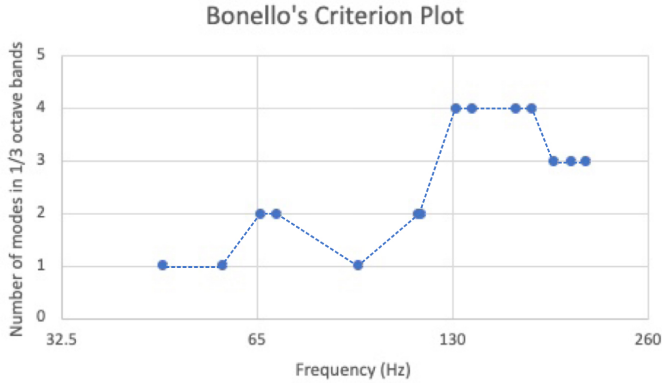


Modal Density

Notice how the circled modes in the $\frac{1}{2}$ octave bands are particularly problematic, these modes correspond to modes 4-6 and 7-11.

6.2 Bonello's Criterion

“To meet Bonello's Criterion, each $\frac{1}{3}$ octave band should have the same or more modes that the previous $\frac{1}{3}$ octave band” (F. Alton Everest, Master Handbook of Acoustics for Musicians). Here is a graph of Weeks's modes in third octave bands to see if the room follows Bonello's Criterion:

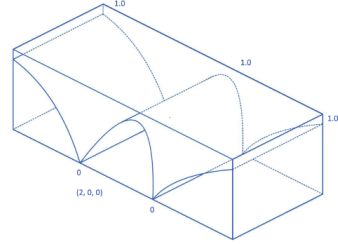
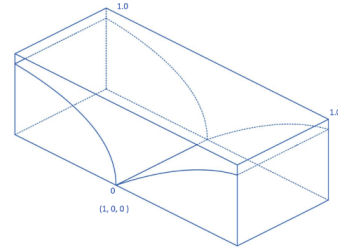


Weeks Bonello's Criterion Test

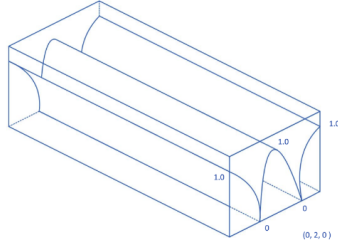
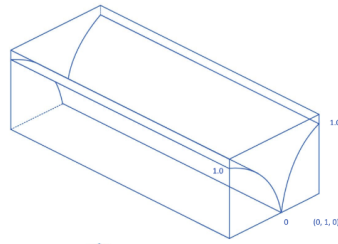
It is clear that this room also does not follow Bonello's Criterion, this is because there are two points in the plot that drop in modes rather than increase in mode number or keep the same number of modes. If we compare this plot to one that the Amroc acoustics website gives it becomes clear that they are quite different. This could be due to the fact that the Amroc website does not take into account the indent of the room nor all the other furniture and objects that are present in the room. This being said, both this plot and the Amroc acoustic plot agree that Weeks 108 does not meet Bonello's Criterion.

7 Visualizing the Axial Modes

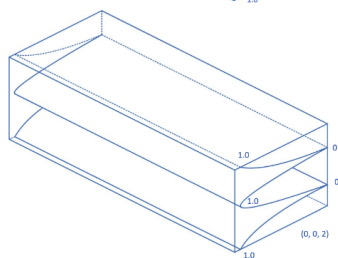
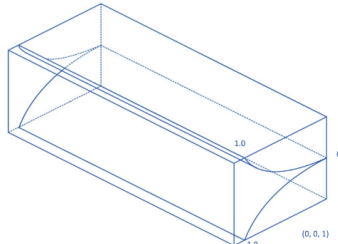
To better get a grasp of what room modes are, and why they are problematic, some diagrams will be drawn. Take notice in the following diagrams how some points are marked 1.0 and some points are marked 0. Areas where 0s are marked represent areas with little to no sound pressure. Areas marked with 1.0s have high sound pressure. Areas of high sound pressure are areas that air is being displaced, while areas with low sound pressure are not being displaced too much. It becomes quite obvious that these nodes can be quite problematic, especially a studio room like Weeks, to have spots where it is hard to hear. Imagine recording a low frequency instrument, such as a bass, or the left hand of the piano, if the microphone is placed in one of these nodes, the sound level could be significantly quieter simply by the placement of the microphone.



Length-Wise Axial Modes



Width-Wise Axial Modes

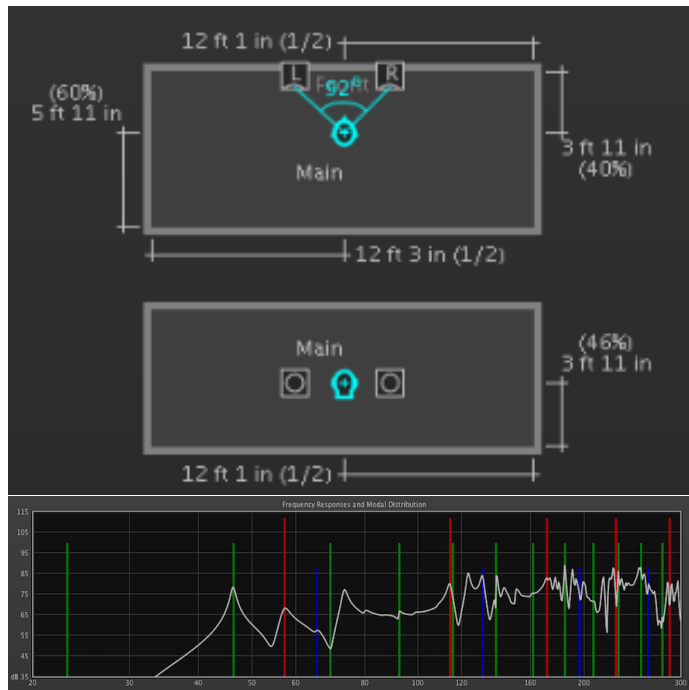


Height-Wise Axial Modes

7.1 Speaker Positions

REW has a room simulator tool that allows users to play around with the listening and speaker position(s). While it was possible to play around with the listening position and

get a better response than the one shown in the plots below, this location was used to consider other non-acoustic factors. This room has a large wooden desk which realistically only has one possible location, against the east wall (the wall that measures 21.66ft). Students also walk from the western door (the door on the wall that measures 24.33 ft) to the eastern door in order to enter the Weeks recording studio. Therefore the listener cannot be in a position of the room that would be blocking walking space. It would also not make sense to have the listener close to the western wall since they would be so far away from the desk that they would not be able to use it; placing the desk against the western wall would block the path to the door on the norther wall. With all this considered, this was the position that had the flattest response.

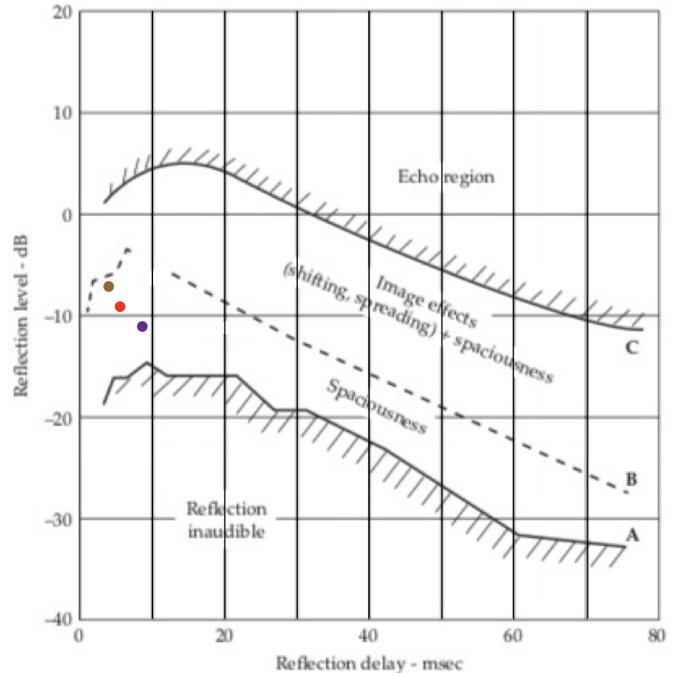


Using these dimensions, I went into to Weeks with a laser and a mirror and calculated the path of the first reflections. The mirror and laser method is simple, by placing the laser in the same location as the speaker and turning it on the laser light will be shown on the wall, the hand-held mirror (placed flush against the wall) can be used to reflect the light back to the listening location. The angle of the laser can be moved, due to the fact that sound will radiate in all directions, to find the right angle that will reflect the light of the laser to the listening position direction. Once everything is set, simply measure the distance from the laser pointer to the wall, and then measure the distance from the wall to the listening position. Here is a chart with all the information: The Reflected Delay's were calculated using this equation:

$$\text{Reflection delay} = \frac{(\text{Reflected path, ft}) - (\text{Direct path, ft})}{1130\text{ft/sec}}$$

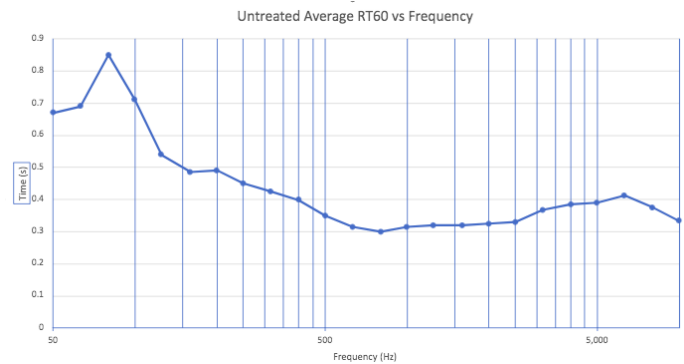
Table 1: Path Characteristics

	Direct	Floor	Ceiling	Rear Wall
Path Length (ft)	3.5	8.33	10	13.42
Direct Reflection (ft)	0	4.83	6.5	0.62
Reflection Level (dB)	-	-7.53	-9.12	-11.68
Reflection Delay	-	4.27	5.75	8.76



Plotting these delays and levels on to a graph gives us a better idea of what type of reflections we are dealing with. All these reflections are in the lower "Spaciousness" section of the graph, which I believe is a good quality of the room.

8 RT60s at 125 Hz, 250 Hz, 500 Hz, and 1000 Hz



Here is the Bass Rise Equation:

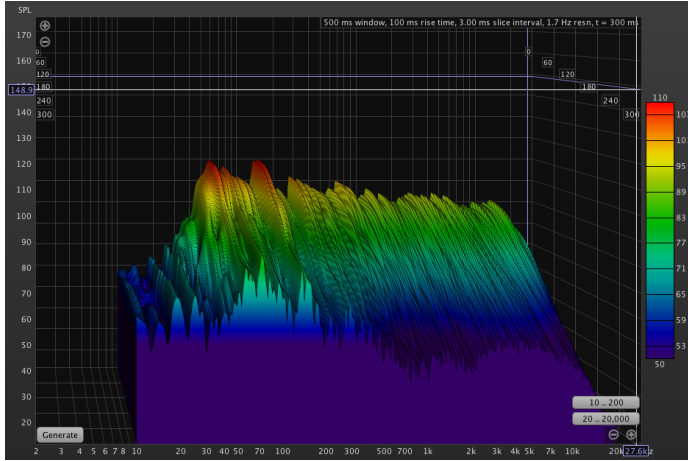
$$\text{BR} = \frac{\text{RT60}_{125} + \text{RT60}_{250}}{\text{RT60}_{500} + \text{RT60}_{1000}}$$

Using the chart we can calculate the bass rise as:

$$\text{BR} = \frac{0.54 + 0.45}{0.35 + 0.315}$$

$$BR = 1.49$$

Generally a room's Bass Rise should be between 1.15 – 1.45, this room's bass rise is a bit too high. Taking a look at the waterfall plot of measurement 25 (The measurement I believe had the closest looking RT60 plot with the average plot above) it can be seen, in accordance to the Modal Density plot and the Bonello's Criterion Plot, that there are loud and prominent (time-wise) frequencies in the range of 50-150 Hz. Some of the mid to high range frequencies are also hanging around. The plan is to treat these lower frequencies first, and by treating the lower frequencies the upper frequencies should also become better.

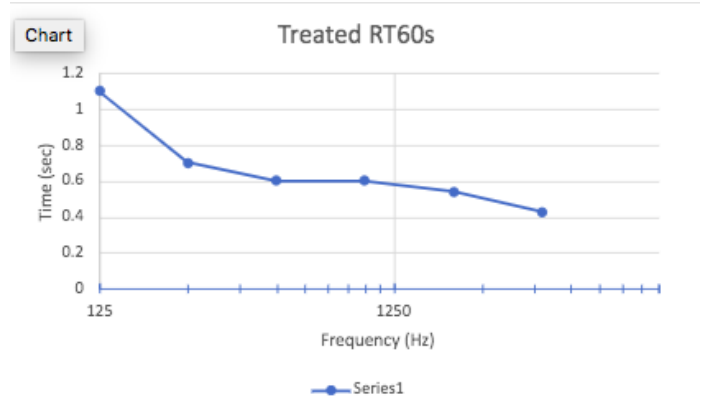


9 Treatment Plan

The frequencies that need to be treated in this space are between 50Hz and 150Hz range. To refresh why these frequencies are the problems in question, refer back to the Axial Modes Plot, the Modal Density Plot, and the Waterfall Plot. Also notice how everything is coming full circle, the Schroeder Cutoff Frequency was calculated to be 148.4 Hz using the REW measurements, very close to 150 Hz, and it was mentioned earlier that it was the frequencies below the Schroeder Frequency that needed to be treated the most. The main focus of the treatment plan is to reduce the modes and the RT₆₀ of those lower frequencies, since they are causing a larger than normal bass rise in such a small room. To treat this it is best to use a bass trap. The next main focus is the hanging reverb time in the middle frequencies. The approach to treat these will be using wall panel absorbers. Here is the treatment plan:

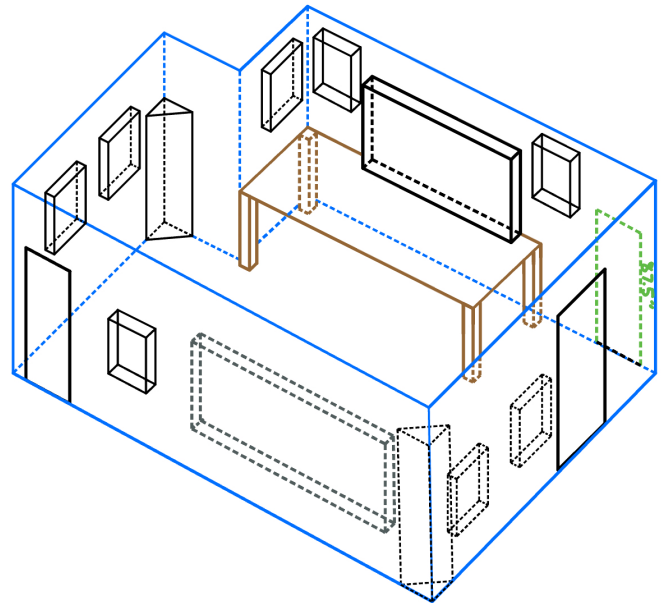
1. 2 GIK Acoustic Tri-Trap Corner Bass Traps. Height 6ft
2. 8 of the 24 in 24 in 2 in GIK Acoustic Spot Panels

In total there will be 75 ft² of wall Absorbers and 60 ft² of Bass Traps. Here is the treated RT₆₀ vs Frequency Plot:



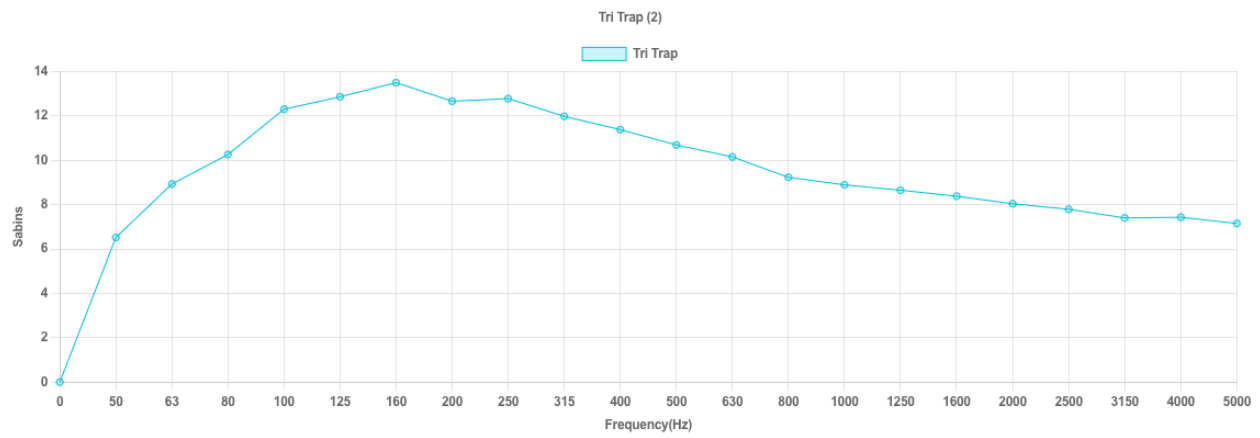
	Treated RT60 Frequencies	Time (sec)
1	125	1.1
2	250	0.7
3	500	0.6
4	1000	0.6
5	2000	0.54
6	4000	0.426

Now you may be asking yourself, why is your treated RT₆₀ worse than the original? Reconsider the original RT₆₀ that was calculated using the Sabine Equation and the RT₆₀ equation. When using this method, the calculated RT₆₀ at 500 Hz was 1.40 seconds (refer back to the Schroeder cutoff frequency section for the explanation as to why the values were so different). Now compare that untreated calculated value, 1.4 seconds, with the new treated value of 0.60 seconds. Clearly the RT₆₀ is better. Here is the floor-plan of where the wall panels and the bass traps would be placed in the room:



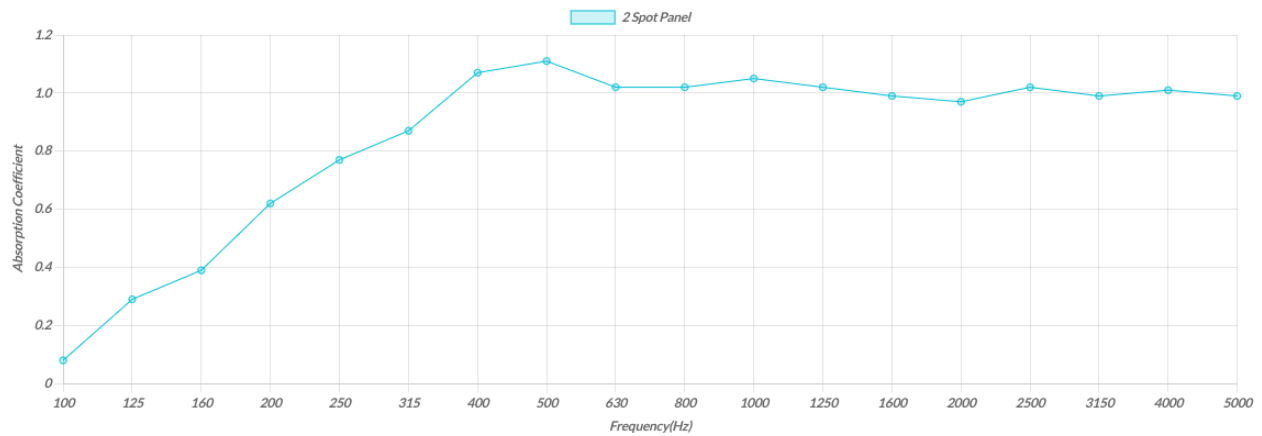
Here are the Specifications of the Tri-trap and Panels:

Technical Specs



Tri-Trap

Technical Specs



Panels